



FARMDIRECT

Direct Farmer-to-Buyer Marketplace with AI Price Intelligence & Hybrid Logistics

Problem Statement: Multiple intermediaries reduce farmers' earnings while increasing consumer prices and post-harvest spoilage.

26.6%

Farmer Share of Rupee

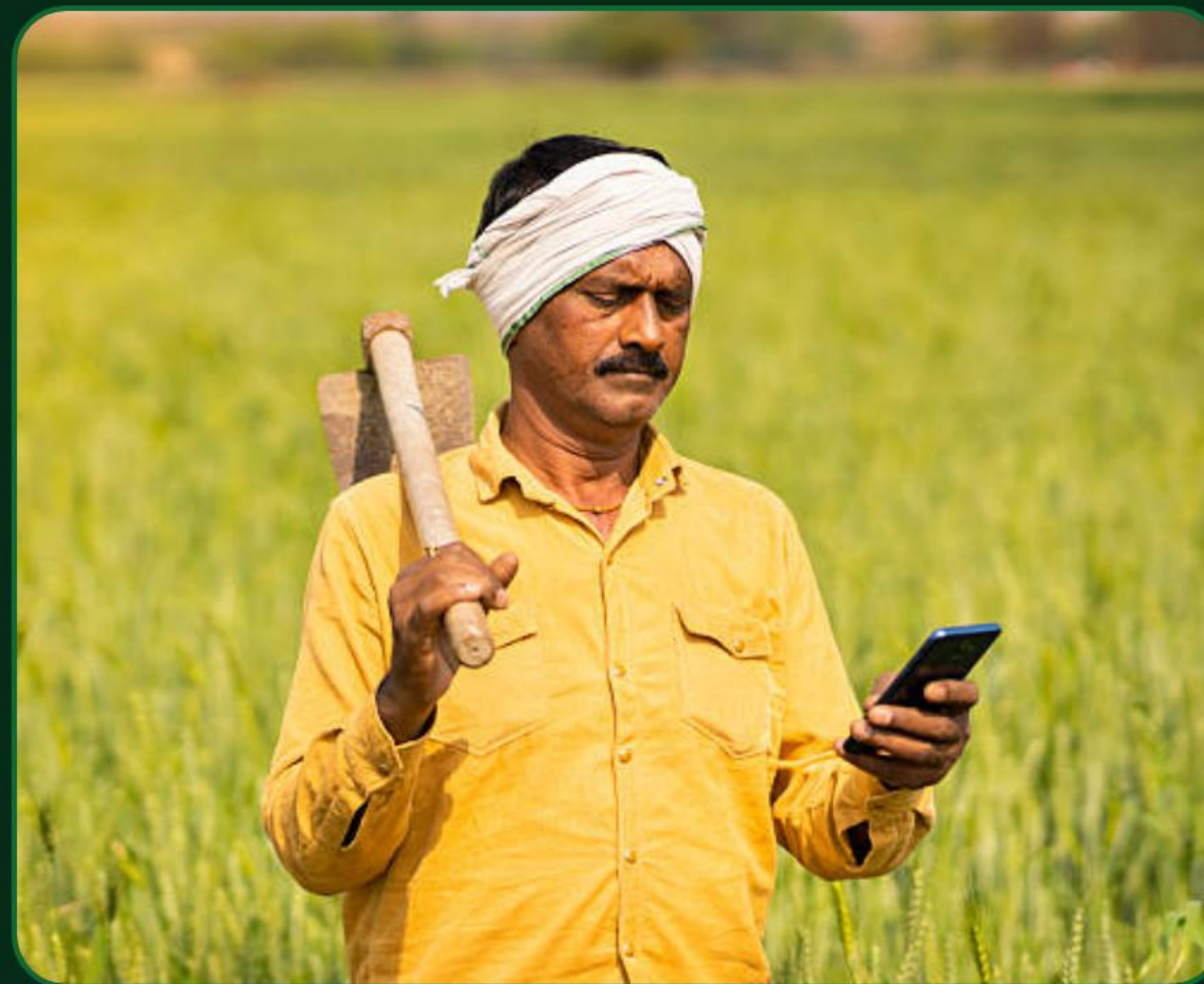
6-15%

Post-Harvest Spoilage

5-6

Intermediary Handoffs Cut

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Problem Worth Solving & Target Ecosystem

Market Analysis

⚠️ Fragmented Supply Chain

Produce moves through 5–6 intermediate hand-offs before reaching end consumers, compounding costs and delays.



26.6%
Farmer Rupee Share

6–15%
Post-Harvest Spoilage

🚜 Small Farmers & FPOs

Pain: Low price transparency.
Value: Multilingual AI voice listing & instant price bounds.

🏠 Kiranas & Retailers

Pain: Unpredictable mandi quality.
Value: Direct scheduled bulk orders with guaranteed quality.

🛒 Urban Households

Pain: High prices for aged produce.
Value: Fresh harvest options with origin transparency.

🚚 Logistics Partners

Pain: Empty return trips.
Value: Clustered batch routes & optimized vehicle loading.

Modular Tech Stack & Core REST APIs

Microservices Spec

Frontend Layer

React Native / Expo with offline-first state caching for low connectivity.

REST Microservices

FastAPI Python microservices with PostgreSQL database & Redis cache.

AI & Routing Engine

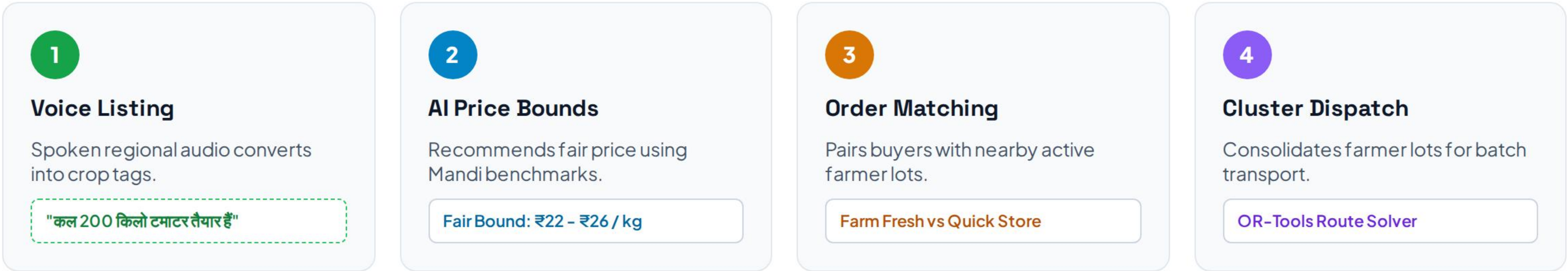
OpenAI Whisper for Speech NLP & Google OR-Tools for batch route solving.

API Specifications & Endpoints

Method	Endpoint	Functionality
POST	/api/v1/voice/parse	Converts Hindi speech to crop metadata
GET	/api/v1/prices/suggest	Fetches Mandi benchmarks & dynamic bounds
POST	/api/v1/orders/create	Executes Farm Fresh or Quick Store order
POST	/api/v1/routes/cluster	Solves optimized cluster pickup routes

End-to-End User Experience & Screen Flow

Miro Flow Spec



Human-in-the-Loop Principle: AI auto-extracts crop quantity and recommends prices, but farmers maintain 100% manual pricing override control before listing publication.

Hybrid Fulfilment Strategy & Execution Roadmap

SLA & Operations

Farm Fresh Track

Harvest and pickup occur post-buyer order placement for maximum freshness with zero inventory risk.

Quick Store Track

High-turnover staples pre-positioned near urban hubs based strictly on AI demand forecasting (1-2 hour delivery).

Execution Timeline & Performance SLAs

- Phase 1: MVP Setup & Voice NLP Months 1-2
- Phase 2: Route Batching & Pilot Test Months 3-4
- Phase 3: Visual Grading & Scale Months 5-6

< 1.8s

Voice Parsing Latency

< 3.0s

Route Solver Speed

Offline Cache

Weak Network Mode